Probability Distributions Handout 2

If integration is required for the problem, show the integration.

1. A product is classified according to the number of defects it contains and the factory that produces it. Let X and Y be the random variables that represent the number of defects per unit and the factory number, respectively. The entries in the table represent the joint probability distribution function of a randomly chosen product.

		Y = 1	Y = 2
X	0	1/8	1/16
	1	1/16	1/16
	2	3/16	1/8
	3	1/8	1/4

   a. Verify that this is indeed a joint probability distribution function.
   b. Find the $P(X + Y > 2)$.
   c. Find the probability that a randomly selected product has less than 2 defects given that it was produced in factory 1.
   d. Find the marginal distribution function of X.
   e. Find the marginal distribution function of Y.
   f. Are X and Y independent? Why or why not?

2. The joint probability distribution function of X and Y is given by

$$f(x,y) = \begin{cases} \frac{6}{7}\left(x^2 + \frac{xy}{2}\right), & 0 < x < 1, \quad 0 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

   a. Verify that this is indeed a joint density function.
   b. Find the marginal distribution function of X.
   c. Find the marginal distribution function of Y.
   d. Find $P(X > Y)$.
   e. Find the $P(X > 0.6 \mid Y = 1.2)$.

3. The joint probability distribution function of X and Y is given by

$$f(x,y) = \begin{cases} xe^{-(x+y)}, & x > 0, \quad y > 0 \\ 0, & \text{otherwise} \end{cases}$$

   a. Find the marginal distribution function of X.
   b. Find the marginal distribution function of Y.
   c. Are X and Y independent? Why or why not?